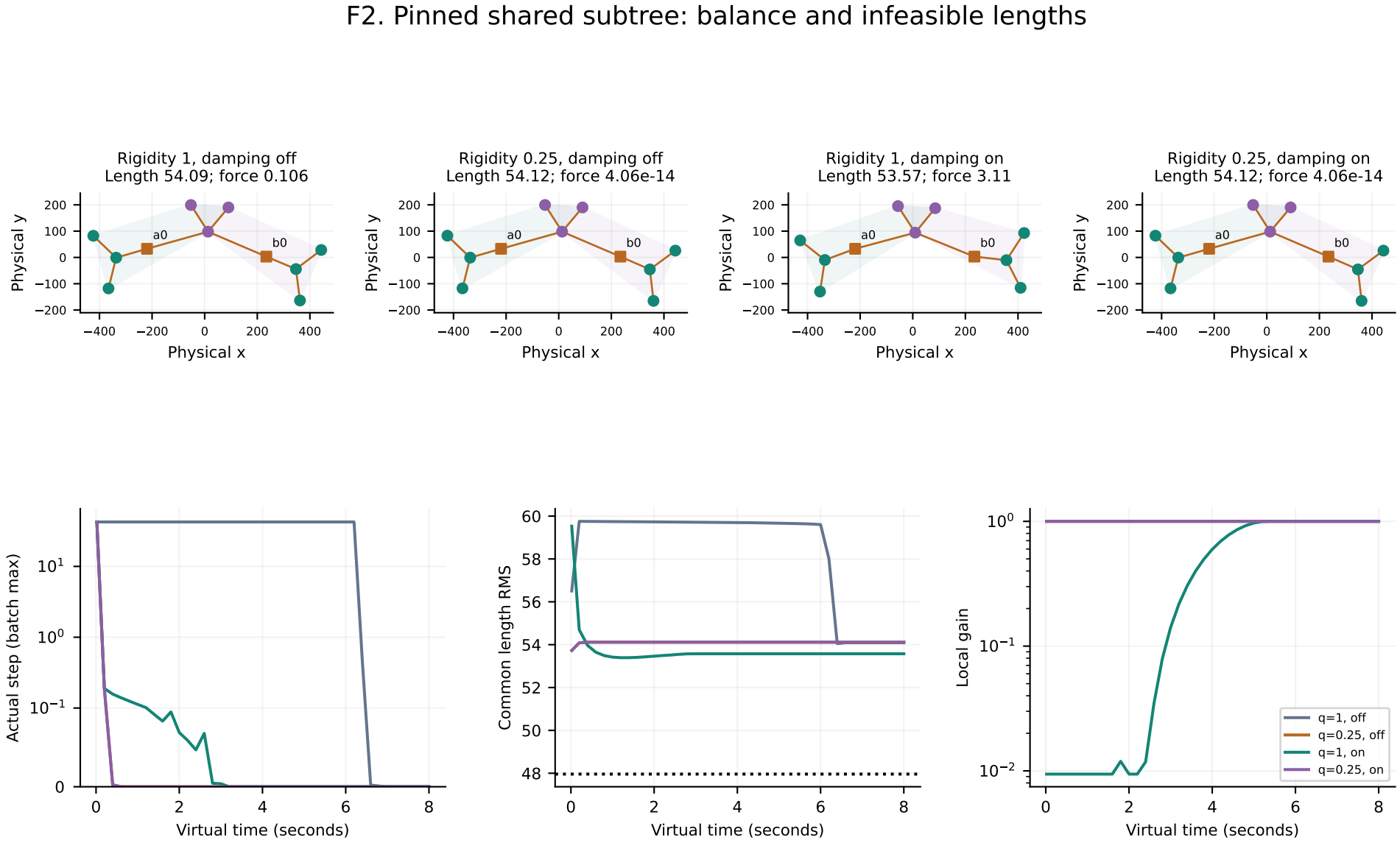

F2. Recorded seed 0x9e3779b9; fixed roots are squares, shared members purple. All final panels share physical axes. Shading denotes membership; orange lines are active constraints. The pinned root distance prevents all targets being satisfied. A near-zero undamped force can coexist with length error. Traces sample selected batches; the force labels are separate final-coordinate probes.